Vahid Hakimian

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EDUCATION

Doctor of Philosophy (Ph.D.), Electrical Power Engineering Aug. 2023 – Present

University of Calgary, Alberta, Canada

- GPA: 4.00/4.00
- Supervisors: Dr. Mostafa Farrokhabadi, and Dr. Hamidreza Zareipour
- Research experience: Transactive markets for net-zero electrical distribution systems, using optimization and Pyomo features.

Master of Science (M.Sc.), Electrical Power Engineering Sep. 2016 – Sep. 2018

Tarbiat Modares University, Tehran, Iran

- GPA: 3.81/4.00
- Supervisor: Dr. Hossein Seifi
- Research experience: Robust dynamic state estimation for power systems with synchronized phasor measurement units, modeling with MATLAB and GAMS.

Bachelor of Science (B.Sc.), Electrical Power Engineering Sep. 2011 – Sep. 2015

University of Tehran, Tehran, Iran

- GPA: 3.33/4.00
- Supervisor: Dr. Mahdi Davarpanah
- Research experience: Grounding transformer transient analysis caused by earth fault clearance in a subtransmission substation and proposing appropriate solutions, linking MATLAB and PSCAD.

SKILLS

- **Technical:** Industrial automation (PLC), optimization, high-voltage substation design, power transmission line design, electrical installations, power system protection, lighting design.
- **Software:**
  - *Power Systems:* COMSOL, Dialux, PSpice, Multisim, LabVIEW, AutoCAD, PowerWorld, ETAP, EMTP-ATP, CYMGRD, PLS-CADD, MATLAB Simulation, S7-PLCSIM, SIMATIC Manager, DIgSILENT, PSCAD.
  - *Programming:* MATLAB, GAMS, Python.
  - *Microsoft:* Word, Excel, PowerPoint, Visio, Outlook.
- **Professional:** Skilled in strategic problem-solving, conflict resolution, and independent task execution with a strong collaborative mindset. Adept at creative thinking, effective communication, and managing project timelines to ensure success.

RESEARCH INTERESTS

- Electrical Power Distribution Grids
- Power System Operation

ENGINEERING EXPERIENCES

High Voltage Substation Design Engineer Nov. 2018 – Jun. 2023

- Organization: Khorasan Power Engineering Consultant (Moniran) Company, Mashhad, Iran.
- Accomplishments: Designed over 20 high-voltage substations across eastern Iran, prepared comprehensive tender documents, supervised project progress on-site to ensure quality, and verified factory tests of high-voltage equipment for compliance and reliability.

Engineering Consultant May 2017 – Aug. 2017

- Organization: Iran Power System Engineering Research Center (IPSERC), Tehran, Iran.
- Accomplishments: Research and analysis of the quality of the primary frequency control in the national grid.

PROFESSIONAL SERVICE

University of Calgary Graduate Students' Association Jun. 2026 – Present

- *Vice Chair, Academic Events Committee:* Support the planning and execution of graduate academic events, including the flagship Peer Beyond Conference and monthly initiatives, in collaboration with the committee Chair and Executive Academic Officer.

IEEE University of Calgary Student Branch Nov. 2023 – Present

- *PES Chair & Associate Co-Chair of Events:* Coordinate technical talks, industry panels, and workshops, leading initiatives that connect students, researchers, and professionals in renewable energy, smart grids, and AI applications.

Journal Peer Review Jun. 2023 – Present

- *Reviewer, IEEE Access:* Review manuscripts focusing on energy systems, smart grids, and emerging power technologies.

University of Tehran, University Music Concerts May 2015

- *Executive Group Member:* Supported planning and delivery of cultural events by coordinating logistics and performer engagement.

University of Tehran, Research & Technology Exhibition Committee Mar. 2015

- *Student Committee Member:* Managed exhibition logistics, prepared publications, and coordinated interviews to highlight student research achievements.

IEEE University of Tehran Student Branch Sep. 2014 – Jun. 2015

- *Volunteer, Research & Publications:* Contributed to Jarian Magazine and supported scientific publication efforts, promoting knowledge dissemination among students.

PUBLICATIONS

Journal Papers

[1] **V. Hakimian**, M. Farrokhabadi, and H. Zareipour, “Data-driven distribution locational marginal emission estimation in congested distribution network operation,” manuscript under preparation, intended for submission to *IEEE Trans. Power Systems*, anticipated submission: Sep. 2026.

[2] H. Yao, **V. Hakimian**, M. Farrokhabadi, and H. Zareipour, “Emission-aware transactive demand response,” intended for submission to *IEEE Trans. Smart Grid*, anticipated submission: Aug. 2026.

[3] S. E. H. Kakolaki, **V. Hakimian**, J. Sadeh, and E. Rakhshani, “Comprehensive study on transformer fault detection via frequency response analysis,” *IEEE Access*, vol. 11, pp. 81852–81881, 2023, https://doi.org/10.1109/ACCESS.2023.3300378.

[4] **V. Hakimian** and H. Seifi, “An optimization-based robust dynamic state estimation for power systems with synchronized phasor measurement units, involving disturbance rejection,” *Int. Trans. Electr. Energy Syst.*, vol. 2023, 2023, https://doi.org/10.1155/2023/1820478.

Conference Papers

[5] H. Yao, **V. Hakimian**, M. Farrokhabadi, and H. Zareipour, “Emission-aware operation of standalone electrical energy storage systems,” *Proc. 59th Hawaii Int. Conf. Syst. Sci. (HICSS)*, Maui, HI, USA, Jan. 2026, p. 3263, https://doi.org/10.24251/HICSS.2026.390.

[6] **V. Hakimian** and M. Davarpanah, “Grounding transformer transient analysis caused by earth fault clearance in a subtransmission substation and proposing appropriate solutions,” *17th Int. Conf. Prot. Autom. Power Syst. IPAPS 2023*, pp. 1–7, 2023, https://doi.org/10.1109/IPAPS58344.2023.10123315.

Poster Presentations

[7] **V. Hakimian**, M. Farrokhabadi, and H. Zareipour, “Nodal marginal emission estimation for emission-aware prosumer participation in distribution networks,” *IEEE PES General Meeting (GM)*, Montreal, Canada, Jul. 2026.

[8] H. Yao, **V. Hakimian**, M. Farrokhabadi, and H. Zareipour, “Emission-aware transactive markets for electrical distribution systems,” *WIRED Symposium*, Aug. 2025.

[9] H. Yao, **V. Hakimian**, M. Farrokhabadi, and H. Zareipour, “Carbon accounting and credit arbitrage in electrical distribution system,” *Alberta Power Industry Consortium (APIC)*, May 2025.

Book Contributions

[10] H. Seifi and H. Delkhosh, *Model Validation for Power System Frequency Analysis*, ser. SpringerBriefs in Energy. Singapore: Springer Nature, 2019. Acknowledged as technical reviewer in the Preface. https://doi.org/10.1007/978-981-13-2980-7.

Technical Reports

[11] **V. Hakimian** and R. Zamordi, “Technical instructions for validating high-voltage substation design reports and compliance manuals,” Khorasan Engineering Consulting Company, Substation Design Technical Group Manual, 2022.

[12] H. Seifi, M. Yousefian, H. Delkhosh, and **V. Hakimian**, “Research and analysis on the quality of primary frequency control in the national power grid: Requirements for monitoring and supervision of primary frequency control performance and proposing implementation methods,” Iran Grid Management Company (IGMC), National Power System Technical Report No. 09, Tehran, Iran, 2017.

HONORS AND AWARDS

Graduate Student Experience Initiative Award, GSA, University of Calgary.	Mar. 2026
Best Paper Award Nomination, 59th Hawaii Int. Conf. Syst. Sci. (HICSS), for "Emission-Aware Operation of Standalone Electrical Energy Storage Systems."	Jan. 2026
Alberta Graduate Excellence Scholarship, University of Calgary.	Sep. 2025
Teaching Assistant Award, ENEL 487, ESE Dept., University of Calgary.	Jun. 2025
Graduate Representative Council Initiative Award, GSA, University of Calgary.	Apr. 2025
ESE Graduate Research Award, Schulich School of Engineering, University of Calgary.	Mar. 2024 & Mar. 2025
Ph.D. Admission with Four-Year Full Funding Scholarship, University of Calgary.	Aug. 2023
Ranked 3rd in GPA, among Electrical Power Systems students, Tarbiat Modares University.	Aug. 2018
Ranked 76th out of ∼29,000 (top 0.26%) in Iran's national M.Sc. entrance exam for Electrical Engineering.	May 2016
Top 30% GPA among ∼120 students in B.Sc. Electrical Engineering, University of Tehran.	Aug. 2015
Ranked 426th out of ∼250,000 (top 0.17%) in Iran's national entrance exam for Mathematics and Physics.	Jun. 2011